



Sponsorship App - Complete Project Explanation

Theoretical & Technical Overview

This document provides complete theoretical explanation of the Sponsorship App project including problem statement, objectives, workflow, technical stack, and complete data flow from frontend to backend.



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1 What is Sponsorship App?

Project Overview

The **Sponsorship App** is a digital platform that connects **Brands** with **Influencers** for marketing campaigns. It's a marketplace where brands can post campaigns and influencers can apply to them, get paid for their promotional work.

Simple Analogy:

Think of it like Fiverr or Upwork but specifically for influencer marketing:

-  **Brands (Sellers of campaigns):** Post campaigns with budget
-  **Influencers (Buyers of work):** Apply for campaigns they like
-  **Platform (Middleman):** Manages approvals, payments, ratings
-  **Admin (Moderator):** Monitors platform, handles disputes

Example:

- Nike wants 50 Instagram posts about their new shoes
- Posts campaign "Nike Summer Collection" with \$50,000 budget
- 200 influencers apply
- Nike selects 50 best fits
- Influencers post content
- Nike pays influencers \$1000 each

Core Entities

Entity	What They Do	Example
Brand	Company with products to promote, posts campaigns	Nike, Adidas, Coca-Cola
Influencer	Person with large following, applies to campaigns	Fashion blogger, Tech YouTuber, Travel Photographer
Campaign	Marketing project posted by brand with budget and requirements	"Instagram Summer Campaign - \$5000 budget"
Sponsorship Request	Application when influencer applies to campaign	John applies to Nike Campaign (pending → approved → completed)
Payment	Transaction after influencer completes work	\$1000 payment for John's Instagram posts
Rating	Review/feedback after campaign completes	Brand rates influencer 5/5 stars

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Problem Statement

Problems in Current Influencer Marketing

✗ Problem 1: Lack of Centralized Platform

Currently, brands and influencers:

- Find each other through scattered social media channels
- Use email for communication (disorganized)
- Have no standardized contracts or agreements
- No proper record-keeping of deals

Result: Time-consuming, error-prone, unprofessional

✗ Problem 2: Trust & Verification Issues

- Brands can't verify influencer's real follower count
- Influencers can't verify brand's legitimacy or payment capability
- No verification of work completion
- Dispute resolution is difficult

Result: Fraud, scams, non-payment, fake followers

✗ Problem 3: Payment & Contract Problems

- No escrow system → Brands don't pay or influencers don't deliver
- Payment tracking is manual (check transfers, PayPal, etc)
- No audit trail for disputes
- Tax documentation is missing

Result: Financial disputes, legal issues

✗ **Problem 4: Performance Metrics Missing**

- No way to track campaign performance
- No analytics on influencer quality
- No ratings system for accountability
- Brands can't make data-driven decisions

Result: Wasted marketing budgets

✗ **Problem 5: Scalability Issues**

- Manual processes don't scale to thousands of campaigns
- Finding right influencers takes time
- Matching campaigns to influencers is inefficient
- Admin workload is massive

Result: Business can't grow

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How We Overcome the Problem

Solution Architecture

✓ Solution 1: Centralized Digital Platform

- Single website/app for all brand-influencer interactions
- Standardized process for posting, applying, approving
- All communication in one place
- Automatic record-keeping in database

Impact: Professional, organized, efficient

✓ Solution 2: Identity Verification & Trust System

- Email-based authentication (verified login)
- User profile with verified information
- Rating/review system (accountability)
- Work verification before payment
- Transparent history of all transactions

Impact: Trust & transparency

✓ Solution 3: Secure Payment System

- Integrated payment processing
- Escrow-like system (payment held until approval)
- Automatic payment after work verification
- Audit trail for all transactions

- Invoice generation for taxes

Impact: Safe, traceable payments

✓ **Solution 4: Analytics & Metrics Dashboard**

- Admin dashboard: Campaign statistics, platform health
- Brand dashboard: Campaign performance, influencer quality
- Influencer dashboard: Earnings, ratings, opportunities
- Filters and search: Find best-fit campaign/influencer

Impact: Data-driven decisions

✓ **Solution 5: Automated Workflow**

- Automatic notifications (email alerts)
- Standardized steps: Apply → Approve → Verify → Pay
- Auto-calculations (payment amounts, fees)
- Admin moderation (flag suspicious activities)

Impact: Scale to thousands of campaigns



Project Objectives

Short-term Objectives (MVP - Minimum Viable Product)

Objective	Description	Status
User Authentication	Secure login/registration for all roles (ADMIN, BRAND, INFLUENCER)	 Implemented
Campaign Management	Brands can create, edit, delete campaigns with specifications	 Implemented
Campaign Discovery	Influencers can browse campaigns with filters (platform, budget, dates)	 Implemented
Application System	Influencers apply to campaigns, brands approve/reject	 Implemented
Work Submission	Influencers submit proof of work (screenshots, links)	 Implemented
Payment Processing	Automated payment after work verification	 Implemented
Rating System	Both parties can rate each other for accountability	 Implemented
Admin Dashboard	Admin can monitor platform, view statistics, handle disputes	 Implemented

Long-term Objectives (Future Enhancements)

- **Real Audience Analytics:** Verify influencer's actual follower engagement using APIs
 - **AI Matching:** ML algorithm to match best influencers to campaigns automatically
 - **Contract Automation:** Smart contracts for payment release based on conditions
 - **Multi-currency Support:** Accept payments in different currencies
 - **Mobile App:** Native iOS/Android apps for better user experience
 - **Content Calendar Integration:** Connect with Instagram, TikTok, YouTube APIs
 - **Advanced Reporting:** Campaign ROI calculation, influencer performance metrics
 - **Dispute Resolution:** Built-in arbitration system
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Project Workflow & Use Cases

Main Workflow

1. Brand creates account & logs in
Posts campaign: "Instagram Summer Campaign"
Budget: \$5000, Dates: June-August

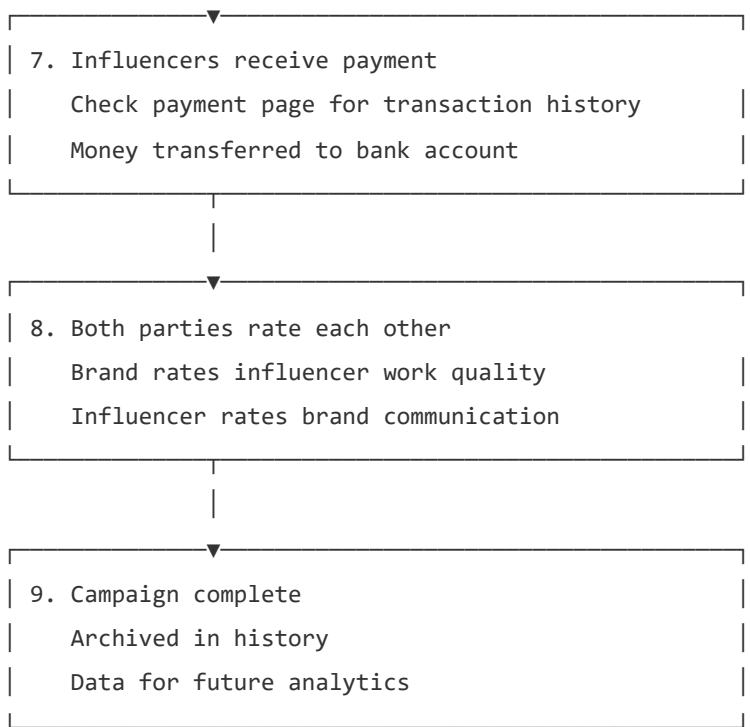
2. Influencer sees campaign in browse page
Clicks "Apply for this campaign"
Submits proposal: "I have 50K followers..."

3. Brand receives notification
Reviews applications from 50+ influencers
Decides to approve 10 influencers

4. Selected influencers get approved
Must post content by end date
Budget split: \$500 per influencer

5. Influencers create posts & submit proof
Post screenshot/link from Instagram
System records submission timestamp

6. Brand verifies work
Clicks "Approve" for each completed post
Payment triggered automatically





Technology Stack

Frontend Technology

Technology	Purpose	Why Chosen
Angular 16+	Frontend framework	Enterprise-grade, type-safe, component-based, large community
TypeScript	Language with type checking	Catches errors at compile time, better IDE support
Angular Material	Pre-built UI components	Professional look, follows Material Design spec, responsive
RxJS	Reactive programming library	Handles asynchronous operations, event management
SCSS	Styling language	Variables, mixins, better CSS organization
npm/Node.js	Package manager & runtime	Dependency management, build automation

Backend Technology

Technology	Purpose	Why Chosen
Spring Boot 3.2.5	Backend framework	Enterprise Java, rapid development, battle-tested, large ecosystem

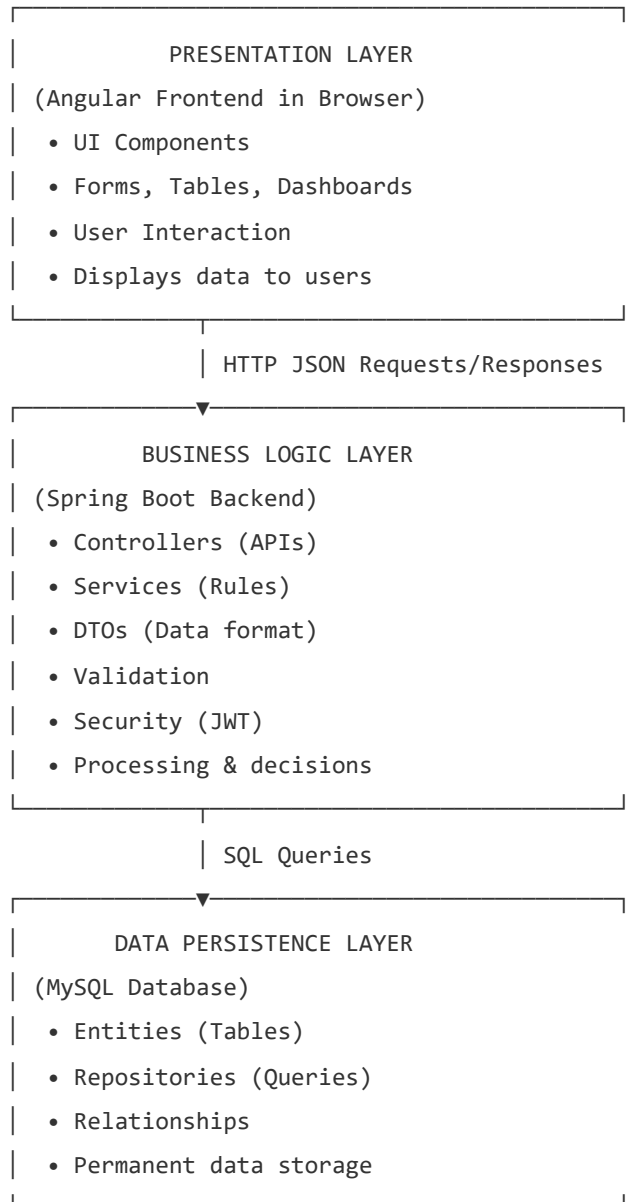
Java 17	Programming language	Latest LTS version, performance improvements, modern features
Spring Security	Authentication & authorization	Enterprise security framework, well-tested, flexible
JWT Tokens	Stateless authentication	Scalable, microservices-friendly, standard in industry
Spring Data JPA	Database ORM	Reduces boilerplate, Hibernate underneath, easy queries
Lombok	Code generation	@Data for getters/setters, reduces boilerplate 70%
MySQL 8	Relational database	Reliable, proven, good performance, free & open source
Maven	Build tool	Dependency management, consistent build process

DevOps & Infrastructure

Tool	Purpose	Current Use
Git	Version control	Track code changes, collaboration
REST API	Server communication	HTTP endpoints for frontend-backend interaction
CORS (Cross-Origin)	Security configuration	Allow frontend (localhost:4200) to call backend (localhost:8080)
LocalHost	Development environment	Frontend: 4200, Backend: 8080

7 Architecture Overview

Three-Tier Architecture



Component Layer Breakdown

Frontend (Angular)

Backend (Spring Boot)

Components → Services → Models →
Guards → Interceptors

Controllers → Services → Repositories
→ Entities → Database



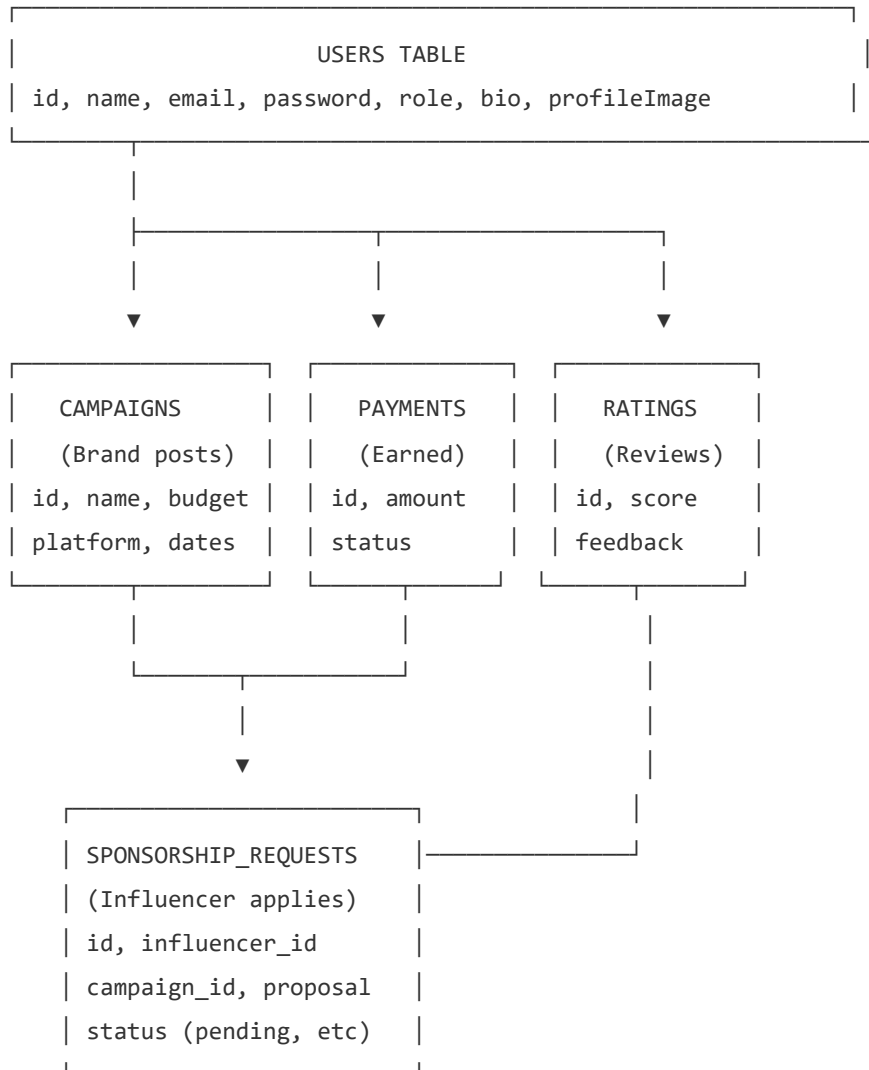
Three User Roles

Role	Characteristics	Permissions	Dashboard Shows
ADMIN	Platform moderator	View all users, suspend accounts, view platform stats, manage disputes	Total users, campaigns, payments, platform health
BRAND	Company posting campaigns	Create campaigns, view applications, approve/reject influencers, process payments	My campaigns, active applications, total spent
INFLUENCER	Content creator	Browse campaigns, apply, submit work, view earnings, get payments	Available campaigns, my applications, earnings

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Data Models & Relationships

Database Schema



Key Relationships

- **User → Campaign (One-to-Many)**: One brand can post many campaigns
- **Campaign → SponsorshipRequest (One-to-Many)**: One campaign gets many applications

- **Influencer → SponsorshipRequest (One-to-Many):** One influencer applies to many campaigns
 - **SponsorshipRequest → Payment (One-to-One):** Each completed application gets one payment
 - **User → Rating (One-to-Many):** Users give and receive ratings
-

10 Complete Data Flow: Login Button to Backend

Step-by-Step Login Flow with Complete Details

STEP 1-5: FRONTEND USER ACTION

Step 1: User opens app in browser

- └ Browser loads index.html
- └ Angular initializes
- └ AppComponent loads
- └ Router redirects to /login (no token in localStorage)

Step 2: LoginComponent displays form

- └ Two input fields: email, password
- └ Click LOGIN button

Step 3: User enters credentials

- └ Email: john@gmail.com
- └ Password: password123
- └ Clicks LOGIN button

Step 4: LoginComponent creates LoginRequest DTO

```
loginRequest = {  
  email: "john@gmail.com",  
  password: "password123"  
}
```

Step 5: AuthService converts to JSON

```
JSON String: {"email":"john@gmail.com","password":"password123"}
```

STEP 6-10: HTTP TRANSMISSION

Step 6: HTTP Request created

Method: POST

URL: http://localhost:8080/api/auth/login

Headers: {

Content-Type: application/json

}

Body: {"email":"john@gmail.com","password":"password123"}

Step 7: AuthInterceptor intercepts request

✓ Checks for token (none yet, so doesn't add)

✓ Logs request

Passes request to HTTP handler

Step 8: Network transmission occurs

Browser sends HTTP POST to backend server

Data travels over network/localhost

Step 9: Backend receives request

Spring Boot server receives HTTP packet

Parses headers and body

Step 10: Routing to Controller

Spring checks URL pattern: /api/auth/login

Matches @PostMapping("/login") in AuthController

Routes request to AuthController.login() method

STEP 11-20: BACKEND PROCESSING

Step 11: AuthController.login() method invoked

@PostMapping("/login")

public ResponseEntity<> login(@RequestBody LoginRequest request) {

Step 12: Spring deserializes JSON → LoginRequest DTO

Jackson library converts JSON string to Java object:

request.email = "john@gmail.com"

request.password = "password123"

Step 13: Validate LoginRequest DTO

Spring checks validation annotations:

@Email: "john@gmail.com" → Valid format ✓

@NotBlank: "password123" → Not empty ✓

Result: Validation passes

Step 14: Call AuthService.login(request)

```
authService.login(request);
```

Step 15: AuthService finds user by email

```
UserRepository.findByEmail("john@gmail.com")
```

SQL Query: SELECT * FROM users WHERE email = 'john@gmail.com'

Result: User entity returned with id=1, name="John Doe", etc

Step 16: Authenticate password

Compare entered password with database hash:

```
BCryptPasswordEncoder.matches(  
    "password123",           ← User entered  
    "$2a$10$hash..."      ← Database stored  
)
```

Result: Passwords match ✓

Step 17: Generate JWT token

```
JwtTokenProvider.generateToken(authentication)
```

- Creates token with email as subject
- Sets expiration: 24 hours from now
- Signs with secret key
- Result: "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."

Step 18: Set SecurityContext

```
SecurityContextHolder.setAuthentication(authentication)
```

✓ User is now authenticated in current request

Step 19: Create AuthResponse DTO

```
AuthResponse response = new AuthResponse(  
    token: "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",  
    id: 1,  
    name: "John Doe",  
    email: "john@gmail.com",  
    role: "INFLUENCER",  
    type: "Bearer"  
);
```

Note: Password is NOT included!

Step 20: AuthController returns response

```
return ResponseEntity.ok(response);
```

HTTP Status: 200 OK

STEP 21-25: RESPONSE SERIALIZATION & TRANSMISSION

Step 21: Spring serializes AuthResponse → JSON

Jackson converts Java object to JSON string:

```
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "type": "Bearer",
  "id": 1,
  "name": "John Doe",
  "email": "john@gmail.com",
  "role": "INFLUENCER"
}
```

Step 22: Create HTTP Response

Status: 200 OK

Headers: {

Content-Type: application/json

}

Body: [JSON from step 21]

Step 23: Backend sends response

HTTP response packet transmitted over network

Step 24: Frontend receives response

Angular HttpClient receives 200 OK response

STEP 26-30: FRONTEND RESPONSE HANDLING

Step 25: AuthInterceptor intercepts response

Status code is 200 (not 401)

✓ No error

✓ Passes response to service

Step 26: AuthService.login() receives response

Angular deserializes JSON → AuthResponse object:

```
response = {
  token: "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  id: 1,
  name: "John Doe",
  email: "john@gmail.com",
  role: "INFLUENCER"
}
```

Step 27: AuthService handles response

handleAuthResponse(response):

- localStorage.setItem('token', response.token)
- localStorage.setItem('user', JSON.stringify({...}))

- `currentUserSubject.next(user)`

Step 28: `LocalStorage` now contains:

Key: token

Value: "eyJhbGciOiJ..."

Key: user

Value: {"id":1,"name":"John Doe",...}

Step 29: `LoginComponent` receives response

Observable subscription completes

Check if login successful

Step 30: Navigate to dashboard

`this.router.navigate(['/dashboard/influencer'])`

OR

`this.router.navigate(['/dashboard/brand'])`

(based on role)

STEP 31-33: DASHBOARD LOADS

Step 31: Router navigates to `/dashboard/influencer`

`InfluencerDashboardComponent` loads

Step 32: `AuthGuard` triggered on route change

Checks: `authService.isLoggedIn()`

Local storage has token?

YES ✓ → Proceed

Step 33: `RoleGuard` triggered

Checks: `currentUser.role === 'INFLUENCER'`

YES ✓ → Load dashboard

Step 34: Dashboard displays

Shows:

- Available campaigns to apply
- My current applications
- My earnings
- Performance metrics

RESULT: ✓ USER SUCCESSFULLY LOGGED IN!



1 1 Complete Data Flow: Campaign Creation

BRAND USER CREATES CAMPAIGN

1. FRONTEND: Brand user fills form
 - Campaign name: "Nike Summer Collection"
 - Platform: Instagram, TikTok
 - Budget: \$10,000
 - Start date: 2024-06-01
 - End date: 2024-08-31
2. FRONTEND: Creates CampaignRequest DTO

```
{
  name: "Nike Summer Collection",
  platform: "Instagram,TikTok",
  budget: 10000.0,
  startDate: "2024-06-01",
  endDate: "2024-08-31"
}
```
3. HTTP: POST to /api/campaigns
 - Authorization: Bearer [JWT token from login]
 - Body: CampaignRequest (as JSON)
4. BACKEND: AuthInterceptor on frontend adds token
 - Authorization header includes JWT
5. BACKEND: AuthInterceptor (on backend) validates token
 - Decodes token
 - Extracts email: "brand@nike.com"
 - Finds user: Nike Brand Company
6. BACKEND: CampaignController receives request
 - @PostMapping("/api/campaigns")
 - @RequestBody CampaignRequest request
7. BACKEND: Deserialization
 - JSON → CampaignRequest object

8. BACKEND: Validation

```
@NotBlank name: "Nike Summer Collection" ✓  
@Positive budget: 10000.0 ✓
```

9. BACKEND: CampaignService.createCampaign()

- Get current brand user from SecurityContext
- Create Campaign ENTITY (not DTO):
campaign.setName("Nike Summer Collection")
campaign.setBrand(currentBrand)
campaign.setStatus(ACTIVE)
campaign.setCreatedAt(now)

10. BACKEND: Save Campaign to database

```
campaignRepository.save(campaign)  
✓ Database generates ID = 15  
✓ INSERT into campaigns table
```

11. BACKEND: Create notification for brand

```
"Your campaign created successfully"
```

12. BACKEND: Return response

```
ApiResponse {  
    success: true,  
    message: "Campaign created",  
    data: Campaign entity  
}
```

13. BACKEND: Serialize to JSON

```
Response sent as JSON
```

14. FRONTEND: Receive response

```
Angular deserializes response
```

15. FRONTEND: Check success

```
if (response.success) {  
    showMessage("Campaign created!")  
    navigate('/campaigns')  
}
```

16. FRONTEND: Redirect to campaigns list

17. FRONTEND: CampaignListComponent loads

```
Fetches all campaigns from /api/campaigns  
New campaign (id:15) appears in list!
```

✓ CAMPAIGN CREATED AND VISIBLE!



1 2 Complete Data Flow: Influencer Application

INFLUENCER APPLIES TO CAMPAIGN

1. FRONTEND: Influencer browses campaigns
Sees "Nike Summer Collection" (id: 15)
Budget: \$10,000
Dates: June-August
2. FRONTEND: Clicks "APPLY FOR THIS CAMPAIGN"
3. FRONTEND: Modal opens asking for proposal
Text area: "I have 100K Instagram followers in fashion niche"
4. FRONTEND: Creates SponsorshipApplicationRequest DTO

```
{
  campaignId: 15,
  proposal: "I have 100K Instagram followers..."
}
```
5. HTTP: POST to /api/influencer/apply
Authorization: Bearer [JWT token]
Body: SponsorshipApplicationRequest
6. BACKEND: AuthInterceptor validates JWT
Extracts influencer user: "influencer@example.com"
7. BACKEND: SponsorshipController receives request
campaignId: 15 exists in database? YES ✓
8. BACKEND: SponsorshipService.applyToCampaign()
 - Get current influencer from SecurityContext
 - Get campaign with id: 15 from database
 - Create SponsorshipRequest ENTITY:

```
sponsorship.setInfluencer(currentInfluencer)
sponsorship.setCampaign(campaign)
sponsorship.setProposal("I have 100K followers...")
sponsorship.setStatus(PENDING)
```

```
sponsorship.setAppliedAt(now)
```

9. BACKEND: Save to database

```
sponsorshipRepository.save(sponsorship)
```

✓ Database generates ID = 42

✓ INSERT into sponsorship_requests

10. BACKEND: Send notification to NIKE BRAND

```
Get campaign.getBrand() = Nike Company
```

```
NotificationService.send(
```

```
    brandUser,
```

```
    "New application from influencer"
```

```
)
```

11. BACKEND: Return response

```
ApiResponse {
```

```
    success: true,
```

```
    message: "Application submitted",
```

```
    data: sponsorship details
```

```
}
```

12. FRONTEND: Show success message

```
"Your application has been submitted!"
```

13. FRONTEND: Redirect to /sponsorship-requests

14. FRONTEND: Show influencer's application status

```
Campaign: Nike Summer Collection
```

```
Status: PENDING
```

```
Submitted: Today
```

BRAND SEES APPLICATION

15. FRONTEND: Brand gets notification

```
Notification bell displays "1 new application"
```

16. FRONTEND: Brand clicks notification → dashboard

17. FRONTEND: BrandDashboardComponent

```
Shows new applications to brand's campaigns
```

18. FRONTEND: Brand sees "Nike Summer Collection"

```
Applicant: [Influencer name]
```

```
Followers: 100K
```

```
Proposal: "I have 100K followers..."
```

```
Buttons: APPROVE | REJECT
```

19. FRONTEND: Brand clicks APPROVE
20. HTTP: POST to /api/brand/approve/{applicationId}
applicationId: 42
21. BACKEND: SponsorshipService.approveApplication(42)
Set status to ACCEPTED
Save to database
22. BACKEND: Send notification to influencer
"Your application was approved!"
23. FRONTEND: Influencer sees notification
Application status changed to ACCEPTED
24. FRONTEND: Show "SUBMIT WORK" button
25. FRONTEND: Influencer clicks "SUBMIT WORK"
26. FRONTEND: Modal for work proof
Text: "Posted on Instagram story and feed"
Upload image/screenshot
27. HTTP: POST to /api/influencer/submit-work
sponsorshipId: 42
workDescription: "Posted..."
workProof: [image]
28. BACKEND: Update sponsorship
status = COMPLETED
submittedAt = now
29. BACKEND: Create Payment record
Payment {
 influencer: influencer,
 campaign: campaign,
 amount: 5000 (budget / number of approved)
 status: COMPLETED
}
30. BACKEND: Save Payment to database
31. BACKEND: Send notification
"Payment ready! Check payment page"
32. FRONTEND: Influencer sees payment notification

33. FRONTEND: Navigate to /payments

34. FRONTEND: See payment

Campaign: Nike Summer Collection

Amount: \$5000

Status: COMPLETED

✓ COMPLETE APPLICATION TO PAYMENT WORKFLOW!

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Security Implementation

Security Measures

Security Feature	Implementation	Protects Against
Password Hashing (BCrypt)	Passwords stored as hashes, not plain text	Database breaches exposing passwords
JWT Tokens	Stateless authentication with signed tokens	Unauthorized access
CORS	Cross-origin requests limited to frontend domain	Cross-site request forgery
CSRF Disabled	No CSRF tokens needed (JWT is stateless)	Cross-site attacks (inherent to JWT)
Role-Based Access	RoleGuard checks user role before access	Unauthorized role access
AuthGuard	Checks token before route access	Unauthenticated access
Spring Security	Backend enforces authentication on all endpoints	Direct API access without frontend

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Future Scope & Enhancements

Phase 2 Enhancements

Feature	Description	Benefits	Timeline
Real-time Notifications	WebSocket for instant alerts instead of polling	Users know immediately about updates, better UX	Q1 2025
Mobile App (iOS/Android)	Native apps using React Native or Flutter	Mobile-first audience, push notifications	Q2 2025
AI Influencer Matching	ML algorithm to suggest best-fit influencers per campaign	Better campaign success rate, saves time	Q3 2025
Social Media API Integration	Connect to Instagram, TikTok, YouTube APIs	Auto-verify follower count, track engagement	Q2 2025
Advanced Analytics Dashboard	Campaign ROI, influencer performance metrics	Data-driven business decisions	Q3 2025
Smart Contracts	Blockchain-based automatic payment release	Trustless payments, dispute resolution	Q4 2025
Multi-currency Support	Accept payments in USD, EUR, INR, etc	Global expansion, international influencers	Q2 2025

Content Calendar Integration

Sync with Google Calendar, Notion, Airtable

Better organization, collaboration

Q4 2025

Long-term Vision (2026+)

- **AI-Powered Content Generation:** Auto-generate campaign briefs using GPT
 - **Sentiment Analysis:** Analyze social media sentiment post-campaign
 - **Fraud Detection:** ML to detect fake followers, bots
 - **Marketplace Expansion:** Add products, merchandise from brands
 - **Creator Academy:** Training content for influencers (courses, webinars)
 - **Brand Registry:** Verification system for brands to build trust
 - **Dispute Resolution AI:** Automated arbitration for conflicts
 - **White-label Solution:** Other companies can rebrand and use platform
-



Complete Project Summary

✓ Sponsorship App - Complete Overview

What It Solves:

- ✓ Connects brands and influencers in one platform
- ✓ Eliminates scattered communication
- ✓ Provides trust through verification
- ✓ Automates payments and tracking
- ✓ Scales to thousands of campaigns

How Data Flows (Login Example):

User enters email/password → Frontend creates LoginRequest DTO → HTTP POST to backend → Spring validates → Authenticates user → Generates JWT token → Creates AuthResponse DTO → Sends JSON response → Frontend stores token → Redirects to dashboard

Technology Used:

- ✓ Frontend: Angular, TypeScript, Material UI
- ✓ Backend: Spring Boot, Java 17, Spring Security
- ✓ Database: MySQL 8
- ✓ Authentication: JWT Tokens
- ✓ Architecture: Three-tier, REST API

Future Direction:

- ✓ Mobile apps for iOS/Android
- ✓ AI-powered influencer matching
- ✓ Real-time notifications
- ✓ Blockchain smart contracts
- ✓ Global expansion with multi-currency